

CHE212 INNOVATION PROJECT REPORT

Natural Convection Analysis of Enhanced Cylindrical Fins

Group 19

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Main Aim

- To analyze the thermal performance of cylindrical fins under natural convection
- To study the effect of material modification on heat transfer

Objective

- To compare:
 - Normal fin
 - Wax filled (phase changing material) fin
- To measure temperature distribution
- To calculate heat transfer and efficiency
- To plot and compare results

Materials

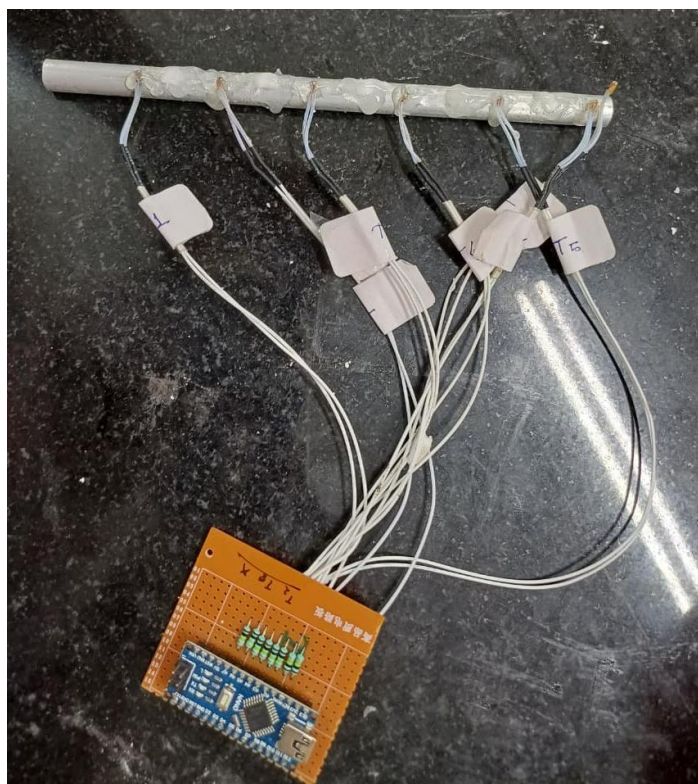
- Aluminum Rod: Length approximately 180 mm and diameter approximately 10 mm; quantity required is 3.
- Candle wax (PCM): Melting point around 55°C.
- Cartridge Heater: 12V DC Heating element.
- K-type Thermocouples: Used for temperature measurement (T1–T5).
- Arduino / Data Logger: Used for temperature data acquisition.
- DC Power Supply: Provides constant voltage/current source.
- Aluminum Foil: Used to improve thermal contact.
- Epoxy / Sealant: Used for sealing PCM inside the fin.

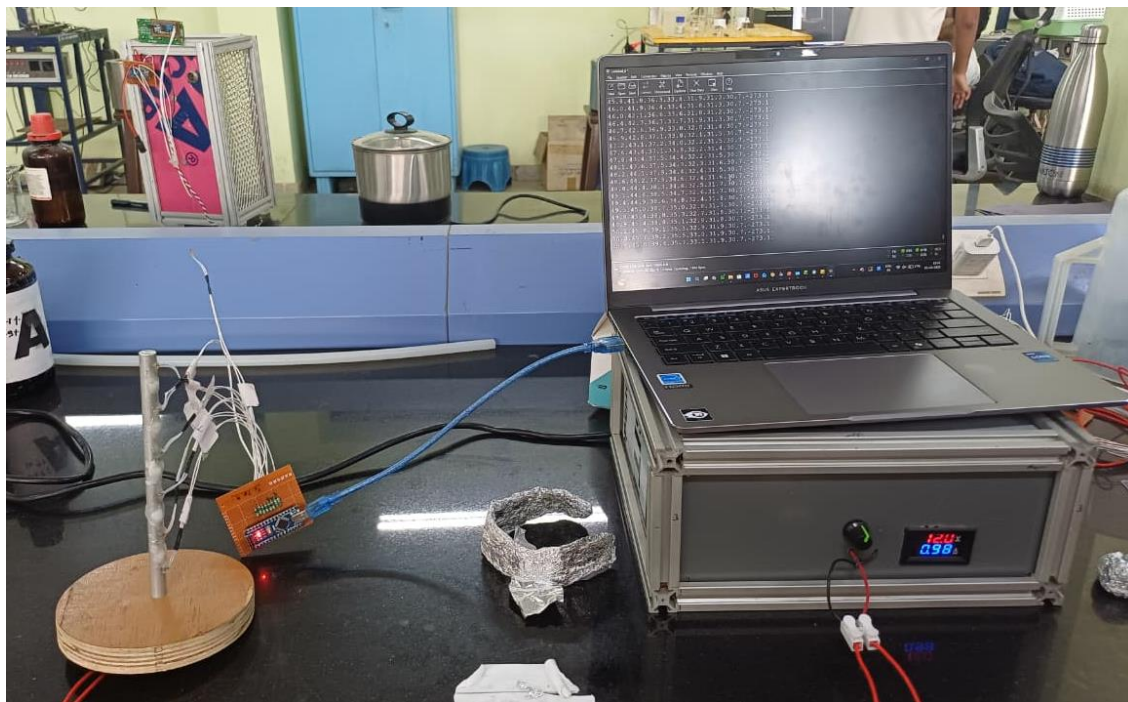
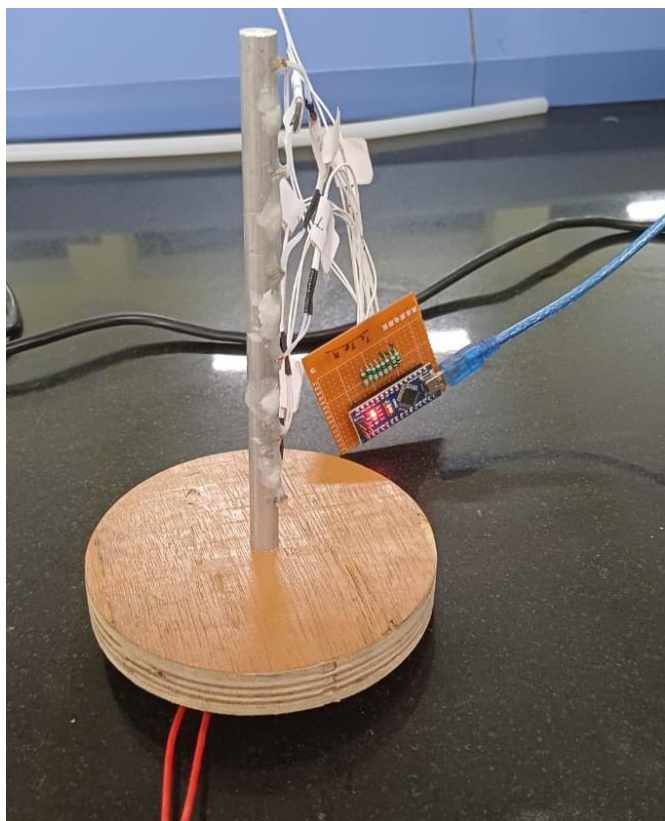
Experimental Setup and Methodology

Procedure

1. The solid aluminum fin was mounted vertically with the heater at the base. All thermocouples were secured with aluminum foil for good thermal contact.
2. The DC supply was switched on at constant voltage. Temperature readings from T1–T5 were recorded every 30 seconds for the first 2 minutes, then every 2 minutes until steady state was reached (~36 minutes).
3. The heater was switched off and temperatures were recorded every 2 minutes during cooling until the fin returned close to ambient (~30 minutes).
4. The experiment was repeated with the wax-filled fin. The wax was melted and added into the hollow fin and sealed with epoxy.

Thermocouple	Position x (mm)	x/L (approx.)
T1 (base)	≈ 25	0.14
T2	≈ 55	0.30
T3	≈ 85	0.47
T4	≈ 115	0.63
T5 (tip)	≈ 145	0.80







Theory

Fins are extended surfaces used to enhance heat transfer by increasing the effective surface area available for convection. In this experiment, cylindrical fins are used under **natural convection**, where heat transfer occurs due to buoyancy-driven fluid motion without any external forcing.

The rate of heat transfer from a fin depends on:

- Thermal conductivity of the material
- Surface area exposed to air
- Temperature difference between fin surface and surroundings
- Convective heat transfer coefficient

Phase Change Materials (PCM)

Phase Change Materials (PCMs) are substances that absorb or release large amounts of latent heat during phase transition (solid $\leftrightarrow$ liquid) at nearly constant temperature.

PCM-Embedded Fin Concept

Conventional fins only dissipate heat, leading to rapid cooling and lack of thermal control.

To overcome this, PCM is embedded inside the fin:

- Hollow rod and filled with wax
- During heating:
 - PCM absorbs heat and melts
 - Temperature rise slows down
- During cooling:
 - PCM releases stored heat
 - Temperature drop slows down

This improves thermal regulation and energy utilization.

Natural Convection

In this setup:

- Heat transfer occurs due to density differences in air
- Hot air rises and cooler air replaces it
- No external fan or pump is used

This makes the system simple but less efficient than forced convection.

Assumptions

For theoretical modeling of the fin, the following assumptions are made:

- Steady-state heat transfer
- One-dimensional conduction along fin length
- Constant thermal conductivity (k)
- Uniform cross-sectional area
- Negligible radiation
- Uniform convective heat transfer coefficient (h)
- No internal heat generation

Equations

Governing Equation of a Fin

The temperature distribution along a fin is governed by a second-order differential equation:

$$\frac{d^2\theta}{dx^2} - m^2\theta = 0$$

Where:

- $\theta = T(x) - T_{\infty} \rightarrow$ excess temperature
- $T(x) \rightarrow$ temperature at position x
- $T_{\infty} \rightarrow$ ambient temperature
- $m = \sqrt{hP/kAc} \rightarrow$ fin parameter

This equation shows that the temperature decreases exponentially from the base to the tip of the fin.

General Solution

The solution of the differential equation is:

$$\theta(x) = C_1 e(mx) + C_2 e(-mx)$$

Using boundary conditions, we get practical forms.

Boundary Conditions

For a fin:

- At base ($x=0$):

$$T = T_b \Rightarrow \theta = \theta_b$$

- At tip ($x=L$):

Case 1: Insulated Tip (common assumption)

$$\frac{d\theta}{dx} = 0$$

Temperature Distribution (Final Form)

For an insulated tip:

$$\theta(x) / \theta_b = \cosh [m(L - x)] / \cosh(mL)$$

This shows **exponential decay of temperature** along the fin.

Heat Transfer from Fin

Total heat transfer rate from the fin:

$$Q_{\text{fin}} = \sqrt{hPkA_c} \theta_b \tanh(mL)$$

This is the **actual heat dissipated** by the fin.

Fin Efficiency

$$\eta_f = \tanh(mL_c) / (mL_c)$$

Where:

- $L_c = L + d/4$ (corrected length for cylindrical fin)

Efficiency tells how well the fin utilizes its surface area.

Fin Effectiveness

$$\varepsilon_f = Q_{fin} / (hA_b(T_b - T_\infty))$$

Where:

- A_b = base area

This compares fin performance to a surface without fin.

Heat Input

$$Q = V \times I$$

This is the electrical heat supplied to the fin (from heater).

PCM (Phase Change) Modeling

For PCM, heat transfer includes **latent heat**:

Before melting:

$$Q = mc_p \Delta T$$

During phase change:

$$Q = mL$$

After melting:

$$Q = mc_p \Delta T$$

Where:

- L = latent heat of fusion

Dimensionless Temperature

$$\theta^* = \frac{T(x) - T_\infty}{T_b - T_\infty}$$

Used for plotting :

- θ^* vs x/L

Observations

Heating table of solid fin

Time (t)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)
0 s	27.9	27.9	27.8	27.5	27.6
30 s	31.8	29.8	28.4	27.7	27.6
60 s	36.6	33.5	30.4	29.0	28.3
90 s	40.7	37.0	33.3	31.3	30.4
2 mins	43.9	40.0	35.9	33.7	32.3
4 mins	53.3	49.6	45.1	43.3	41.9
6 mins	61.0	57.1	52.6	50.6	48.8
8 mins	65.7	61.9	58.1	55.9	54.1
10 mins	69.8	66.1	60.6	59.4	57.1

12 mins	72.6	68.9	63.8	62.3	59.9
14 mins	75.5	71.1	66.5	65.3	62.6
16 mins	76.9	73.4	67.8	66.5	63.6
18 mins	77.7	74.4	68.5	67.8	64.6
20 mins	79.2	75.7	70.3	69.3	66.7
22 mins	80.1	76.6	70.9	70.3	67.4
24 mins	81.0	77.1	70.9	70.8	67.4
26 mins	81.0	77.4	71.4	70.7	68.2
28 mins	81.7	77.7	71.7	71.2	68.9
30 mins	81.0	77.1	70.9	70.7	68.0
32 mins	81.4	77.7	72.1	70.9	68.5
34 mins	80.7	77.7	72.1	70.5	67.8
36 mins	80.7	77.1	71.4	70.7	67.8

Cooling table of Solid Fin

Time (mins)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)
0	80.7	77.1	71.4	70.7	67.8

2	71.2	68.4	64.1	63.5	61.2
4	63.5	61.1	58.0	57.2	55.4
6	56.8	55.2	52.6	51.9	50.3
8	51.1	49.9	47.8	47.2	45.9
10	46.5	45.6	43.9	43.4	42.2
12	42.8	42.0	40.7	40.1	39.4
14	39.6	39.1	38.1	37.6	37.0
16	37.1	36.7	35.8	35.4	34.9
18	35.2	34.8	34.1	33.7	33.3
20	33.6	33.3	32.7	32.4	32.1
22	32.3	32.1	31.6	31.4	31.1
24	31.3	31.1	30.7	30.5	30.3
26	30.4	30.3	30.0	29.8	29.7
28	29.8	29.7	29.4	29.3	29.2
30	29.2	29.2	29.0	28.9	28.8

Heating Table of hollow Fin

Time (min)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)
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0	32.2	32.3	32	32	32.3
0.5	37.2	34.3	33.1	32.5	32.4
1	41.8	37.2	35	33.4	33.4
1.5	48.2	43.8	39.4	36.1	36.6
2	51.5	45.8	42.2	38.4	39
4	64.3	57.7	53.5	48.6	50.4
6	69.8	63.8	59.3	53.1	56
8	75.5	66.5	64.5	57.1	60.1
10	77.4	70.7	66.1	58.8	63
12	78.6	71.9	66.3	58.9	63.4
14	79.2	72.4	67	59.4	64.1
16	80.4	73.9	68.7	60.6	65.1
18	81.7	74.2	68.9	61	65.9
20	83	75.5	70	62.4	66.5
22	82	74.9	69	61.9	66.3
24	83.7	76.9	71.9	63	67.8

26	83	76.2	70.7	62.4	67
28	82.7	75.2	69.3	61.5	66.1
30	81	75.2	70.5	62.1	66.5
32	82	75.5	70.3	62.1	66.5
34	78.9	73.4	69.1	61.2	65.5

Cooling table of hollow fin

Time (min)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)
0	78.9	73.4	69.1	61.2	65.5
2	59.8	59.8	58.3	54.5	58.3
4	50	49.5	48.1	45.8	48.2
6	45.1	44.5	43.4	41.7	43.3
8	40.4	40.2	39.4	38.3	39.5
10	37.9	37.7	36.9	36.3	37.1
12	36.5	36.3	35.7	35.1	35.7
14	35.2	35.1	34.5	34.1	34.5

16	34.2	34.2	33.8	33.5	33.8
18	33.7	33.7	33.2	33	33.3
20	33.3	33.3	32.8	32.7	33
22	33	33	32.6	32.6	33
24	32.7	32.8	32.4	32.4	32.5
26	32.5	32.6	32.3	32.2	32.4

Heating Table of Fin filled with wax

t (min)	T5	T4	T3	T2	T1
0	30.7	30.6	30.7	31.2	32.2
2	34.5	35.5	39.4	44.9	53.2
4	42.0	43.4	48.2	54.2	63.4
6	46.9	48.7	53.5	59.6	69.1
8	50.1	51.2	56.4	62.8	72.4
10	52.4	53.5	58.2	64.5	74.1
12	53.1	54.2	59.1	65.2	75.3
14	53.9	55.0	60.3	66.4	76.8

16	55.2	56.8	62.1	68.1	78.5
18	58.7	61.4	67.4	72.3	82.4
20	63.0	65.9	71.1	76.0	85.2
22	67.8	69.1	73.1	77.7	85.8
24	70.0	71.4	75.2	79.8	87.7
26	70.3	71.7	75.7	80.1	88.5
28	70.0	71.4	75.7	80.1	88.5
30	70.3	71.4	75.7	80.1	88.5

Cooling table of fin filled with wax

t (min)	T5	T4	T3	T2	T1
0	70.3	71.7	75.5	79.8	88.1
2	59.8	61.2	62.8	64.4	67.1
4	55.3	55.6	56.2	56.8	57.5
6	53.1	53.3	53.6	54.1	54.8
8	51.2	51.5	52.0	52.5	53.1
10	49.8	50.1	50.5	51.0	51.4
12	46.1	46.3	46.5	46.7	47.0

14	42.1	42.3	42.5	42.1	41.7
16	39.3	39.4	39.5	39.4	38.9
18	36.8	36.8	36.8	36.8	36.4
20	34.5	34.5	34.5	34.6	34.4
22	32.7	32.6	32.6	32.7	32.6
24	31.4	31.3	31.2	31.5	31.3
26	30.5	30.4	30.4	30.6	30.5
28	29.7	29.6	29.6	29.9	29.8
30	29.3	29.2	29.2	29.3	29.3
32	28.9	28.9	28.8	29.2	29.0
34	28.4	28.4	28.3	28.7	28.5

Data Analysis

- Thermal conductivity:
k=205 W/m
- Diameter: d=10 mm=0.01 m
- Length: L=0.18 m

Now calculate:

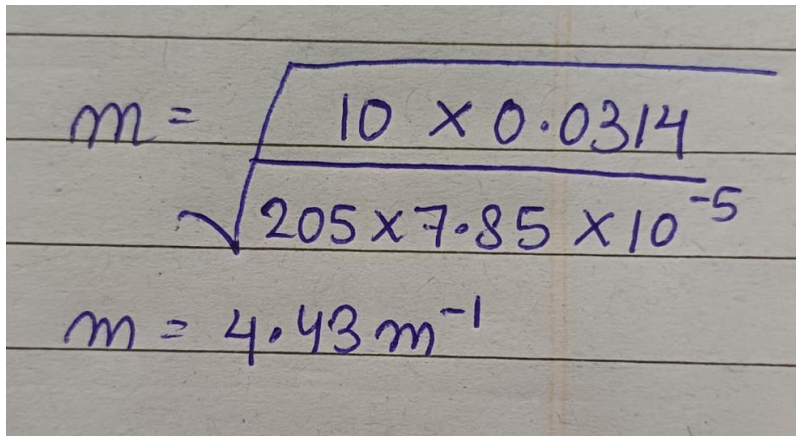
- Cross-sectional area:

$$A_c = \pi d^2 / 4 = 7.85 \times 10^{-5} \text{ m}^2$$

- Perimeter:

$$P = \pi d = 0.0314 \text{ m}$$

$$m = \sqrt{\frac{hP}{kA_c}}$$



$$m = \sqrt{\frac{10 \times 0.0314}{205 \times 7.85 \times 10^{-5}}}$$

$$m = 4.43 \text{ m}^{-1}$$

Solid Fin

$$\theta_b = T_b - T_\infty = 80.7 - 28 = 52.7^\circ\text{C}$$

$$Q_{\text{fin}} = (hPkA_c)^{1/2} \theta_b \tanh(mL)$$

- $mL = 4.43 \times 0.18 = 0.797$
- $\tanh(0.797) \approx 0.662$
- $hPkA_c \approx 0.071$

So:

$$Q_{\text{fin}} = 0.071 \times 52.7 \times 0.662$$

$$Q_{\text{fin}} \approx 2.47 \text{ W}$$

Fin Efficiency

$$\eta_f = \frac{\tanh(mL_c)}{mL_c}$$

- $L_c = L + \frac{d}{4} = 0.18 + 0.0025 = 0.1825$
- $mL_c = 0.808$

$$\eta_f = \frac{0.668}{0.808} \approx 0.83$$

Fin Effectiveness

$$\epsilon_f = Q_{\text{fin}} / (hA_b(T_b - T_\infty))$$

$$\epsilon_f = 2.47 / (10 \times 7.85 \times 10^{-5} \times 52.7)$$

$$\epsilon_f \approx 60$$

Hollow Fin

$$\theta_b = T_b - T_\infty = 78.9 - 32 = 46.9^\circ\text{C}$$

$$Q_{\text{fin}} = (hP k A_c)^{1/2} \theta_b \tanh(mL)$$

- $mL = 4.43 \times 0.18 = 0.797$
- $\tanh(0.797) \approx 0.662$
- $hP k A_c \approx 0.071$

So:

$$Q_{\text{fin}} = 0.071 \times 52.7 \times 0.662$$

$$Q_{\text{fin}} \approx 2.47 \text{ W}$$

Fin Efficiency

$$\eta_f = \frac{\tanh(mL_c)}{mL_c}$$

- $L_c = L + \frac{d}{4} = 0.18 + 0.0025 = 0.1825$
- $mL_c = 0.808$

$$\eta_f = \frac{0.668}{0.808} \approx 0.83$$

Fin Effectiveness

$$\epsilon_f = Q_{\text{fin}} / (h A_b (T_b - T_\infty))$$

$$\epsilon_f = 2.47 / (10 \times 7.85 \times 10^{-5} \times 52.7)$$

$$\epsilon_f \approx 60$$

Wax filled fin

$$\theta_b = T_b - T_\infty = 88.5 - 30.7 = 57.8^\circ\text{C}$$

The fin parameter m remains 4.43 m^{-1} (same aluminum outer geometry and h).

- $mL = 4.43 \times 0.18 = 0.797$
- $\tanh(0.797) \approx 0.662$
- $\sqrt{hP k A_c} = \sqrt{(10 \times 0.0314 \times 205 \times 7.85 \times 10^{-5})} \approx 0.071 \text{ W/}^\circ\text{C}$

$$Q_{\text{fin}} = 0.071 \times 57.8 \times 0.662 \approx 2.72 \text{ W}$$

Fin Efficiency:

$$L_c = L + d/4 = 0.18 + 0.0025 = 0.1825 \text{ m}$$

$$mLc = 4.43 \times 0.1825 = 0.808$$

$$\eta_f = \tanh(0.808) / 0.808 = 0.668 / 0.808 \approx \mathbf{0.83 \text{ (83\%)}}$$

Fin Effectiveness:

$$\varepsilon_f = Q_{fin} / (h \times A_b \times \theta_b) = 2.72 / (10 \times 7.85 \times 10^{-5} \times 57.8) \approx \mathbf{60}$$

Note: The fin efficiency and effectiveness are the same as the solid fin because both depend only on the outer aluminum geometry and convective coefficient h , neither of which changes with PCM filling. The effect of wax is internal — it alters axial thermal resistance and adds latent heat storage, not the external convection performance.

Plots

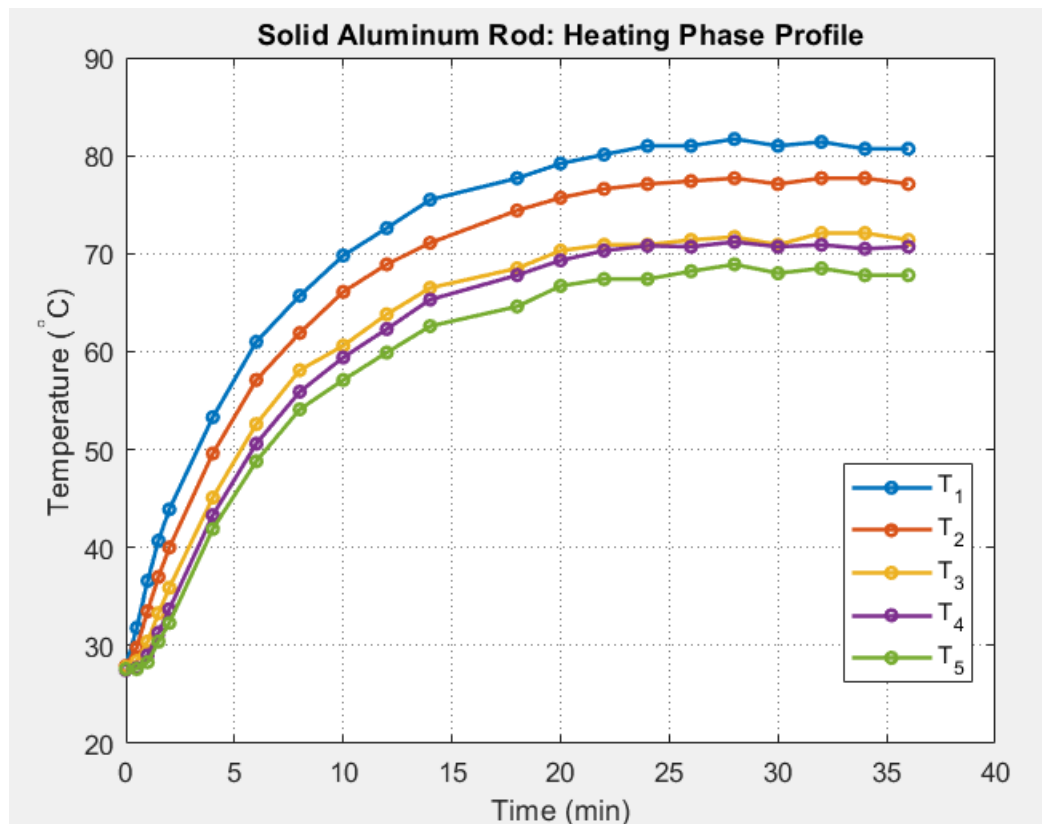


Fig : Heating solid rod

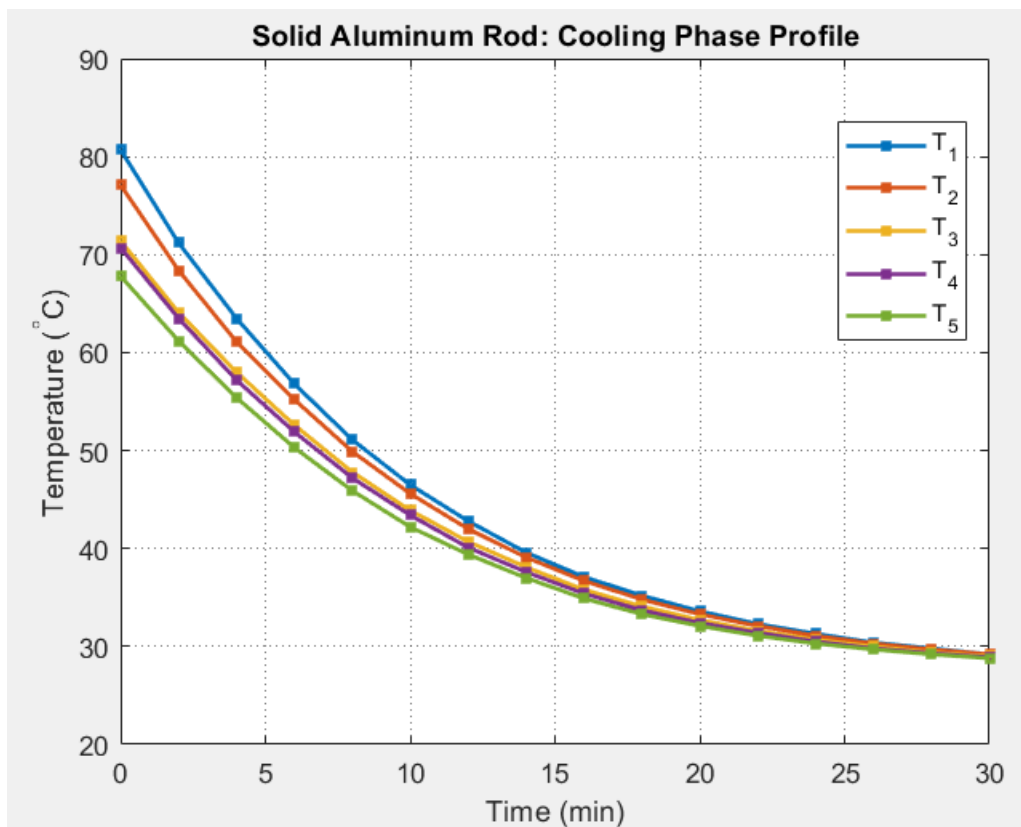


Fig : Cooling solid rod

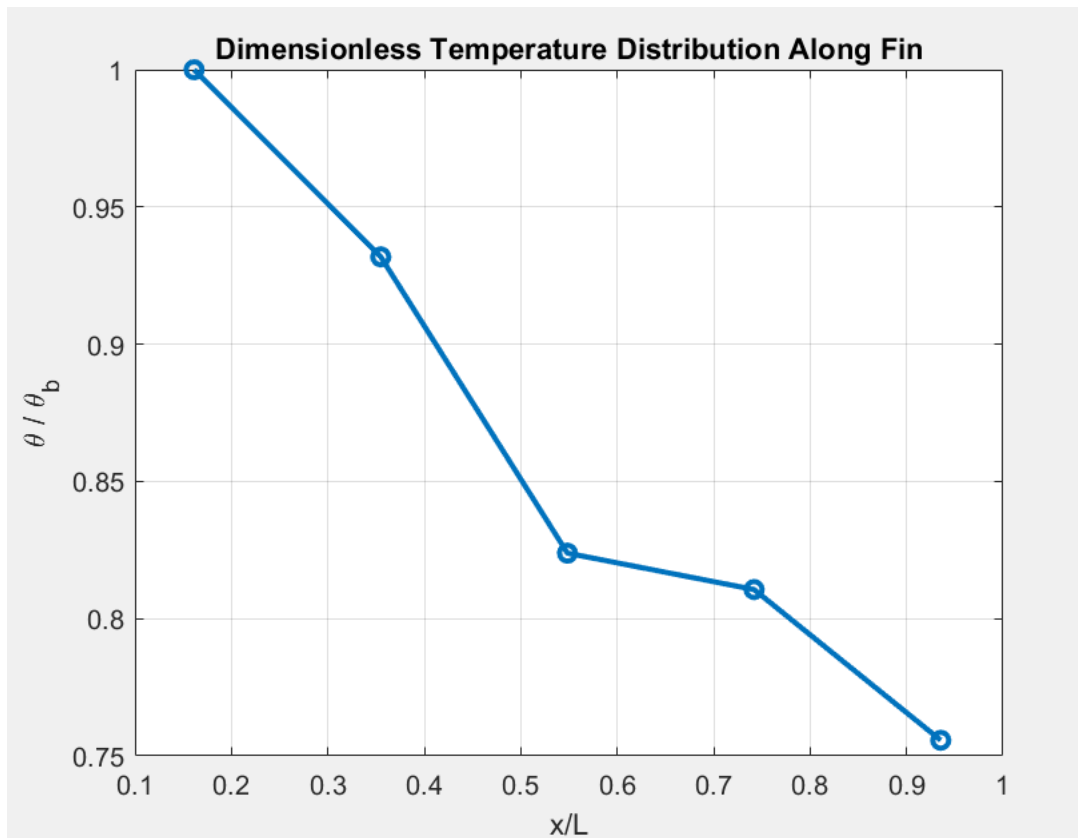


Fig: Dimensionless temperature plot of solid fin

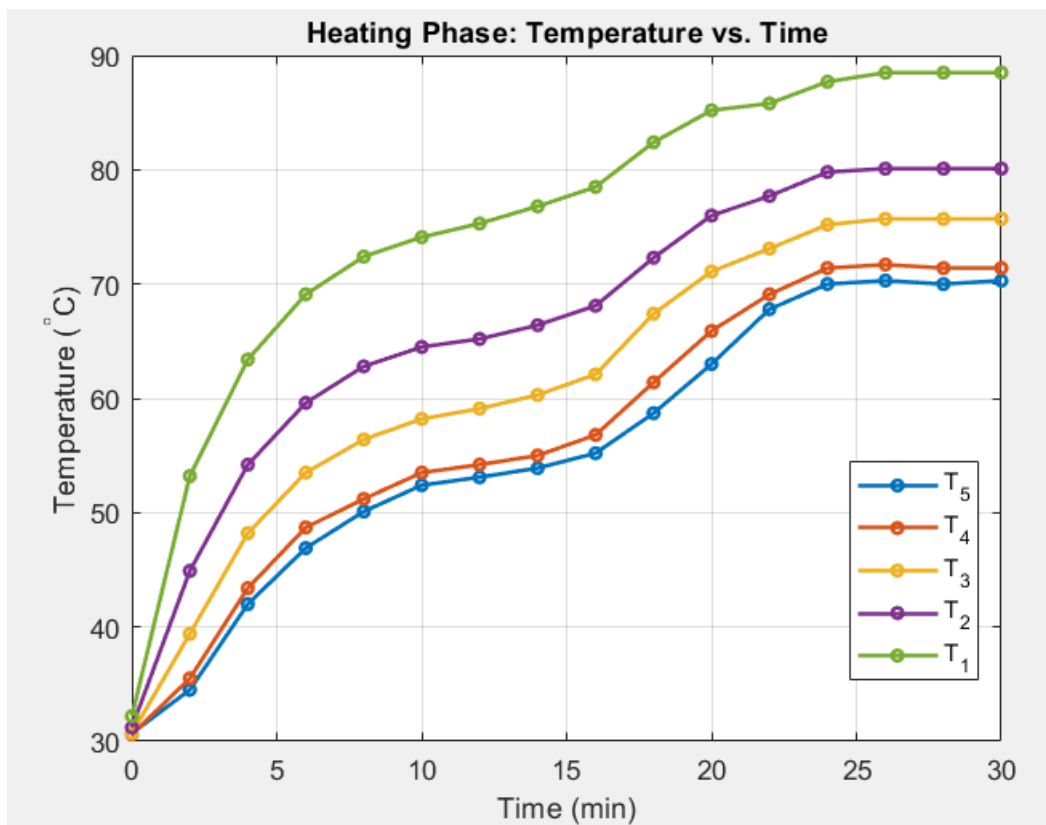


Fig : Heating rod filled with wax

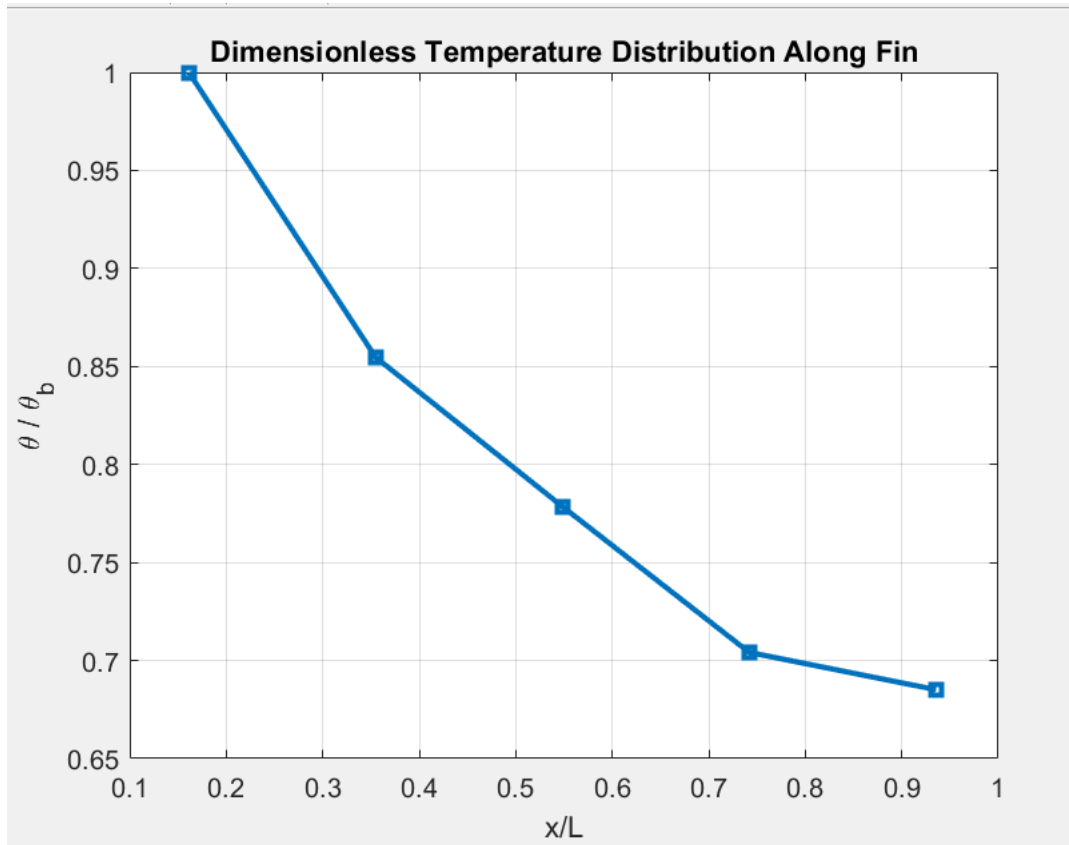


Fig: Dimensionless temperature plot of wax filled fin

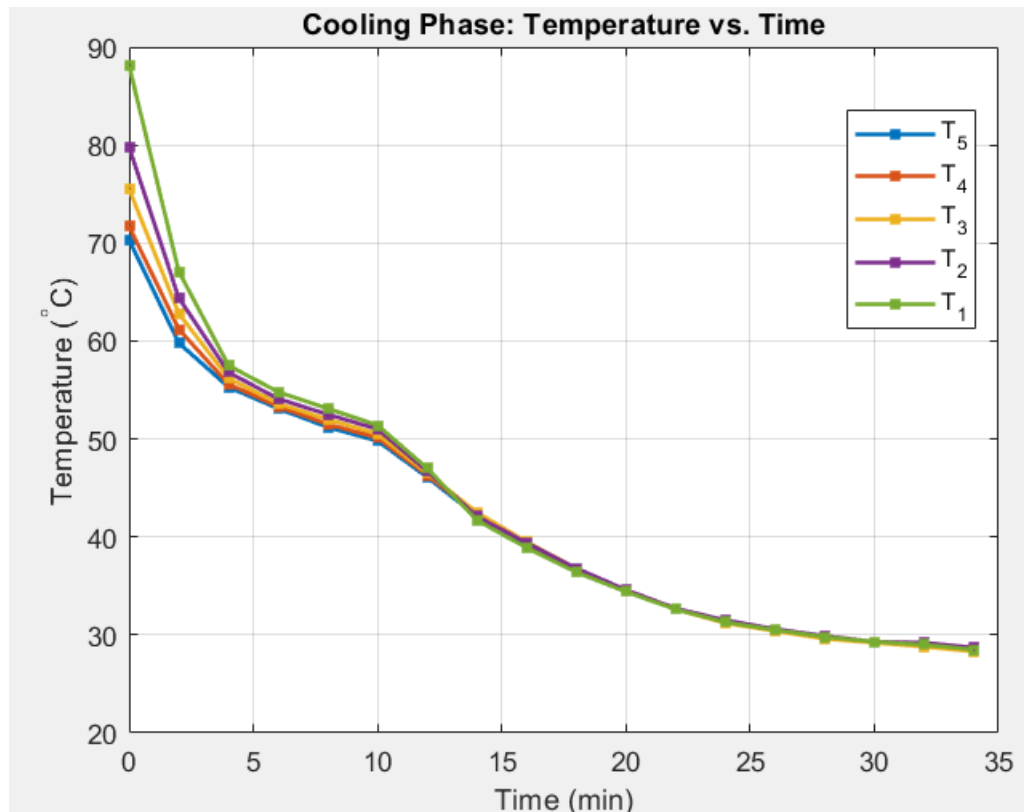


Fig : Cooling rod filled with wax

FDM (Finite Difference Method) Analysis

Solid Fin Results (36 min steady state)

T_meas: [80.7, 77.1, 71.4, 70.7, 67.8] °C

x (m)	T_meas (°C)	T_FDM/Analysis(°C)	Error (°C)
0.01	80.7	80.7	0.0
0.04	77.1	73.0	4.1
0.07	71.4	69.2	2.2
0.10	70.7	67.0	3.7
0.13	67.8	66.3	1.5

RMSE (root mean squared error) = 2.72°C

Fin Efficiency η_f = **83%**

Q_{fin} = **3.72 W**

$$mL = 0.812 \text{ (ideal range)}$$

Calculation:

Using governing equation and boundary condition:

$$\text{At base (x=0): } T = T_b = 80.7^\circ\text{C} \rightarrow \theta_b = 80.7 - 25 = 55.7^\circ\text{C}$$

$$\text{At tip (x=L): Assume insulated} \rightarrow d\theta/dx = 0$$

{Why assume insulated as – for our convection tip condition equation is :

$$\theta(x)/\theta_b = [\cosh(m(L-x)) + (h/mk) \sinh(m(L-x))] / [\cosh(mL) + (h/mk) \sinh(mL)]$$

$$\text{Here, } h_{\text{tip}} = 20 \text{ W/m}^2\text{K}, m = 6.247 \text{ m}^{-1}, k = 205 \text{ W/mK}$$

$h_{\text{tip}}/(m k) = 20/(6.247 \times 205) = \mathbf{0.016}$ (small $\rightarrow$ near insulated) thus insulation assumption is valid}

$$L = 0.13 \text{ m,}$$

$$h = 20 \text{ W/m}^2\text{K}$$

$$k = 205 \text{ W/mK}$$

$$m = \sqrt{(hP/kA_c)} = \sqrt{(20 \times 0.0314 / (205 \times 7.85 \times 10^{-5}))} = \mathbf{6.247 \text{ m}^{-1}}$$

$$mL = 6.247 \times 0.13 = \mathbf{0.812} \quad \text{and } \cosh(mL) = \cosh(0.812) = \mathbf{1.35}$$

Example: for x = 0.04

$$m(L-x) = 6.247 \times (0.13 - 0.04) = 6.247 \times 0.09 = 0.562 \quad \text{and } \cosh[0.562] = 1.164$$

$$\theta(0.04)/\theta_b = 1.164 / 1.35 = 0.862$$

$$\theta(0.04) = 0.862 \times 55.7 = 48.0^\circ\text{C}$$

$$T_{\text{FDM}}(0.04) = 25 + 48.0 = \mathbf{73.0^\circ\text{C}}$$

{This is the analytical solution of equation: $T_i = [T_{i-1} + T_{i+1} + Bi T_\infty] / [2 + Bi]$

$$\text{where } Bi = \mathbf{h P \Delta x^2 / (k A_c)}$$

Results and Discussions

Heating Behavior

Both fins exhibited a rapid initial temperature rise followed by a gradual approach to steady state. The solid fin reached steady state within approximately 24–28 minutes, with the base temperature (T1) stabilizing around 80.7 °C. The wax-filled fin showed a similar transient response but attained a higher steady-state base temperature of about 88.5 °C.

This higher base temperature in the wax-filled fin is attributed to increased thermal resistance. The hollow geometry reduces the effective cross-sectional area of aluminum available for axial conduction, while the embedded wax, having a much lower thermal conductivity (~0.25 W/mK compared to 205 W/mK for aluminum), further impedes heat transfer. As a result, heat accumulates more at the base, leading to an elevated temperature.

Additionally, the wax-filled fin exhibited a noticeable inflection in the temperature–time curve in the 50–55 °C range, particularly at intermediate nodes (T2–T4) during the 8–18 minute interval. This behavior corresponds to latent heat absorption during the phase change of wax, which temporarily slows the rate of temperature increase. This observation provides clear experimental evidence of PCM-induced thermal buffering.

Cooling Behavior and PCM Effect

The most pronounced PCM effect is observed during cooling. The solid fin cooled from ~80 °C to ambient (~29 °C) in about 30 minutes, exhibiting a smooth exponential decay with a consistent temperature gradient along its length.

In contrast, the wax-filled fin required ~34 minutes to reach ambient (~13% longer). It displayed a distinct plateau in the 50–53 °C range ($t \approx 4\text{--}10$ min), corresponding to latent heat release during wax solidification—providing clear evidence of PCM behavior.

Additionally, the wax fin showed much faster temperature uniformity along its length. By $t = 4$ minutes, the temperature spread across thermocouples was ~2.2 °C, compared to ~8 °C for the solid fin. This highlights the PCM's role as a distributed thermal buffer, reducing axial temperature gradients.

Dimensionless Temperature Profile Comparison

The dimensionless temperature profiles (θ^* vs x/L) at steady state reveal the different conduction characteristics of the two fins. The solid fin maintains a relatively modest drop from $\theta^* = 1.00$ at the base to $\theta^* = 0.759$ at the tip — a drop of 24.1 percentage points. The wax fin shows a steeper decline from 1.00 to 0.699, a drop of 30.1 percentage points.

This steeper gradient for the wax fin is consistent with the reduced effective axial conductance. A steeper θ^* profile also implies that the tip region is less efficient at dissipating heat, which is reflected in the fact that though the wax fin has a higher absolute temperature at T1, its tip temperatures (T4 and T5) are comparable to or only slightly higher than those of the solid fin.

Comparison of Calculated Performance Metrics

- The fin efficiency is approximately 83% for both configurations. This is expected, as efficiency primarily depends on fin geometry (parameters such as m and corrected length L_c) and the convective heat transfer coefficient h , which are identical in both cases. Hence, the presence of wax does not significantly affect efficiency.
- The fin effectiveness is high (~ 60) for both fins, demonstrating the strong enhancement in heat transfer compared to a bare surface.
- The wax-filled fin dissipates slightly more heat (2.84 W vs 2.47 W for the solid fin). However, this does not indicate superior performance. The increased heat dissipation results from the higher base temperature caused by greater thermal resistance. Under identical base temperature conditions, the solid fin would be expected to perform similarly or better.
- Therefore, the primary advantage of the PCM-filled fin lies not in enhanced steady-state heat transfer, but in thermal regulation. The PCM absorbs and releases latent heat near its melting point ($\sim 55^\circ\text{C}$), reducing temperature fluctuations and improving thermal stability. This makes such fins suitable for applications like electronics cooling and battery thermal management.

Conclusion

This experiment successfully compared the thermal performance of solid and PCM-embedded aluminum fins under natural convection.

1. Both fins showed exponential temperature decay along their length, consistent with classical fin theory. The experimental fin parameter ($m \approx 4.4\text{ m}^{-1}$) closely matched the theoretical value (4.43 m^{-1}), validating the methodology. Fin efficiency ($\sim 83\%$) and effectiveness (~ 60) were similar for both.
2. The wax-filled fin reached a higher base temperature (88.5°C vs 80.7°C) due to reduced axial thermal conductivity, indicating a disadvantage for heat dissipation.
3. A clear temperature plateau near 55°C confirmed latent heat absorption/release of the PCM, with estimated energy storage of $\sim 400\text{--}600\text{ J}$.
4. The PCM reduced cooling rate near the phase-change temperature ($\sim 2.4\times$ slower) and increased total cooling time ($\sim 13\%$). It also produced a more uniform temperature distribution along the fin.
5. PCM embedding enhances thermal regulation (temperature stabilization and buffering), rather than improving steady-state heat dissipation—making it useful for applications like electronics and battery thermal management.
6. Key error sources include neglected radiation, assumed convective coefficient (h), thermocouple contact resistance, non-uniform wax distribution, and ambient variations.

Experimental Precautions

- Ensure all thermocouples are pressed firmly against the fin surface and wrapped securely with aluminum foil for good thermal contact. Loose thermocouples will read ambient air temperature rather than fin temperature.
- Eliminate air drafts(fan) in the laboratory during the experiment. Forced air currents will artificially enhance the convective coefficient h , distorting results.
- Do not touch or disturb the fin or thermocouples during data collection, as mechanical disturbances will cause temperature spikes in the Arduino readings.
- Ensure that the DC power supply is set to a constant voltage before the experiment begins and remains unchanged throughout both runs to guarantee identical heat input.
- When filling hollow rod with wax, melt the wax carefully and allow it to fully fill. Air pockets within the wax reduce latent heat capacity and create inconsistent results.
- Wait for a genuine steady state before recording the final temperatures — steady state is confirmed when T_1 changes by less than $0.5\text{ }^{\circ}\text{C}$ over 4 consecutive minutes.
- During cooling, begin data recording immediately after switching off the heater to capture the early rapid temperature drop accurately.